Internship
Progress Report



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Project Statement:

Design and Development of Hydraulic Soft Actuator for Single Degree of Freedom

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Abstract:

Soft robotic actuators have emerged as an alternative to conventional rigid actuators due to their compliance, lightweight construction, and ability to safely interact with humans. Among various soft actuators, the McKibben Artificial Muscle (MAM) has gained considerable attention because of its high force-to-weight ratio, simple construction and bio-mimetic contraction mechanism. Most existing McKibben muscles are pneumatically actuated, which limits their application in high-load systems due to the compressibility of air, lower force density, pressure fluctuations, and reduced positional accuracy.

The present research aims to design and develop a fully automatic hydraulic McKibben artificial muscle system capable of actuating a robotic elbow joint while lifting a payload of approximately 5 kg under practical operating conditions. Water or hydraulic oil is proposed as the working fluid to achieve higher force output, improved positional stability, and enhanced actuation efficiency compared to conventional pneumatic systems. In addition, the study investigates the influence of an optimized braided sheath geometry on actuator performance, including force generation, contraction ratio, stress distribution and durability.

The work involves conceptual design, CAD modeling and finite element analysis. The developed system expected to provide improved load-carrying capability, higher stiffness and better operational reliability than conventional pneumatic McKibben actuators

The proposed hydraulic artificial muscle system has potential applications in robotic manipulators. rehabilitation devices, industrial automation, wearable exoskeletons and prosthetic limbs.

Keywords: Hydraulic McKibben Artificial Muscle. Robotic Elbow Joint, Soft Robotics, Hydraulic Actuation, Braided Sheath Optimization, Finite Element Analysis

Introduction:

Soft robotic technologies have transformed the field of robotic actuation by introducing compliant mechanisms capable of interacting safely with humans and unstructured environments. Artificial muscles have become one of the most promising soft actuators because they imitate the contraction mechanism of biological skeletal muscles while maintaining low weight, compact dimensions, and high force generation.

Among different artificial muscle technologies, the McKibben Artificial Muscle is widely recognized for its simple construction and excellent force-to-weight ratio. It consists of an inner elastic bladder surrounded by a braided sleeve that converts radial expansion into axial contraction when pressurized.

Traditionally, McKibben muscles employ compressed air as the working medium. Although pneumatic systems are inexpensive and lightweight, they suffer from several disadvantages. including air compressibility, pressure losses, nonlinear response, hysteresis, limited load-carrying capability, and poor position holding under varying external loads. These limitations become increasingly significant in robotic joints that require accurate positioning and higher payload capacities.

Hydraulic actuation offers an attractive alternative because liquids are nearly incompressible. allowing higher operating pressures, improved stiffness, greater force generation, and superior positional accuracy, Consequently, hydraulic muscles have recently attracted considerable interest for rehabilitation robots, industrial manipulators, underwater robotics and wearable exoskeletons.

However, research on hydraulic muscles is limited, particularly concerning optimized braided sheath configurations, compact hydraulic systems and fully automatic. Most existing studies rely on conventional braided sleeves configurations.
This project therefore focuses on designing a fully hydraulic artificial muscle integrated with a robotic elbow mechanism capable of lifting approximately 5 kg payload. The study includes optimization of the braided sheath, hydraulic pressure analysis, finite element simulation and performance evaluation.

Problem Statement:

To design and develop a single-degree-of-freedom robotic elbow joint using a hydraulic soft actuator in an antagonistic biceps-triceps configuration, capable of lifting a minimum 5 kg load while providing the required joint torque, motion, and structural safety.

Objective:

1. To study the working principle and characteristics of hydraulic McKibben muscle actuators for robotic applications.
2. To design a single-DOF robotic elbow joint using a biceps-triceps type McKibben actuator arrangement for a minimum 5 kg load.
3. To analyze the required joint torque, actuator force, pressure, and braided sheath parameters.
4. To Finalize the suitable for mechanism for rotation of elbow.

Literature Review:

Year	Author(s)	Research Paper / Topic	Publication	Key Finding	Limitation	Relevance
2022	M. G. Antonelli et al.	Design and Characterization of a McKibben Pneumatic Muscle Prototype with an Embedded Capacitive Length Transducer	Machines, MDPI	Developed an embedded capacitive sensor capable of measuring muscle-length changes below 1 mm.	Requires a specialized embedded sensing arrangement.	Useful for future actuator displacement measurement and feedback.
2018/2019	T. Hassan et al.	Finite Element Modeling and Design of a Pneumatic Braided Muscle Actuator with Multifunctional Capabilities	IEEE/ASME Transactions on Mechatronics	Developed a bidirectional braided actuator with more than 40% stroke and demonstrated it in a 1-DOF variable-stiffness joint.	More complex sheath configuration and pneumatic operation.	Highly relevant to our 1-DOF robotic elbow and antagonistic actuator concept.
2018	M. G. Antonelli et al.	An Analytical Formula for Designing McKibben Pneumatic Muscles	International Journal of Mechanical Engineering and Technology	Developed an analytical force formula based on pressure, geometry, and contraction using FEM results.	Valid only within the investigated parameter ranges.	Useful for preliminary sizing of our actuator.
2017	M. G. Antonelli et al.	Numerical Modelling and Experimental Validation of a McKibben Pneumatic Muscle Actuator	Journal of Intelligent Material Systems and Structures	Developed nonlinear FEM considering hyperelastic tube behaviour and validated it experimentally.	Thread friction and initial tube-braid gap were not modelled.	Provides a basis for nonlinear FEM and experimental validation.

Year	Author(s)	Research Paper / Topic	Publication	Key Finding	Limitation	Relevance
2017	A. K. P. Ashwin, A. Ghosal	Mathematical Model for Pressure-Deformation Relationship of Miniaturized McKibben Actuators	iNaCoMM 2017	Developed a pressure-deformation model with less than 10% standard error and predicted the locking angle near 54.7°.	Focused on miniaturized actuators and specific modelling assumptions.	Useful for understanding pressure, deformation, and braid-angle relationships.
2014	T. Nozaki, T. Noritsugu	Motion Analysis of McKibben Type Pneumatic Rubber Artificial Muscle with Finite Element Method	International Journal of Automation Technology	Developed a 3D FEM model and investigated the influence of braid angle on actuator motion.	Friction between tube, fibers, and fiber strands was not directly modelled.	Useful for braid-angle analysis and FEA methodology.
2013	I. da Paz Lopes, W. Scaff, O. Horikawa	Development A Nonconducting Manipulator, McKibben Pneumatic Artificial Muscle	COBEM 2013	Developed and experimentally tested a nonconducting McKibben actuator for a robotic manipulator.	Relatively low force and force reduction was observed, possibly due to leakage.	Useful for actuator fabrication and robotic integration.
2012	K. Iwata, K. Suzumori, S. Wakimoto	A Method of Designing and Fabricating McKibben Muscles Driven by 7 MPa Hydraulics	International Journal of Automation Technology	Developed a hydraulic McKibben muscle and achieved about 8 kN contraction force at 7 MPa. FEM was used to optimize the sleeve angle.	Focused on high-pressure hydraulic operation.	The most directly relevant paper to our hydraulic McKibben actuator and sheath design.

Year	Author(s)	Research Paper / Topic	Publication	Key Finding	Limitation	Relevance
2005	R. Ramasamy et al.	An Application of Finite Element Modelling to Pneumatic Artificial Muscle	American Journal of Applied Sciences	Established FEM-based analysis for studying PAM deformation and force behaviour.	General PAM modelling with limited detailed McKibben-specific nonlinear analysis.	Provides the basic FEM foundation for our actuator analysis.

Table 1: Literature Survey

Literature Survey:

- Previous studies on McKibben artificial muscles have mainly focused on understanding their working principle, mathematical modeling, and the influence of braid geometry on actuator performance. Analytical and experimental studies have established relationships between internal pressure, braid angle, contraction, actuator dimensions, and generated force. The literature also shows that the braided sheath plays a major role in converting radial expansion of the bladder into axial contraction. Finite Element Analysis has therefore been widely used to study deformation, stress distribution, and the effect of braid angle before fabrication.
- Further studies have focused on improving the design and functionality of McKibben actuators. FEM-based investigations have been used for design optimization, while analytical models have been developed for predicting force and pressure-contraction behavior. Researchers have also investigated multifunctional braided muscles capable of bidirectional motion and variable stiffness. These studies demonstrate that actuator geometry, braid configuration, and material properties strongly influence the final performance of the muscle.
- For high-force applications, hydraulic actuation has also been investigated as an alternative to conventional pneumatic McKibben muscles. The hydraulic McKibben study by Iwata et al. demonstrated operation at pressures up to 7 MPa and investigated the design and fabrication of the braided muscle using FEM-based sleeve optimization. The reviewed hydraulic soft-actuator literature further identifies high force capability as an important advantage, while also highlighting challenges such as nonlinear deformation, hysteresis, leakage, and the need for systematic design methods.
- **Research Gap:** Based on the reviewed literature, the present work focuses on applying a hydraulic McKibben muscle in a single-DOF robotic elbow joint using a biceps-triceps type arrangement, designed to lift a minimum 5 kg load. There is practical validation can also be done.

Derivation:

Derivation of Pressure and Force Relation:

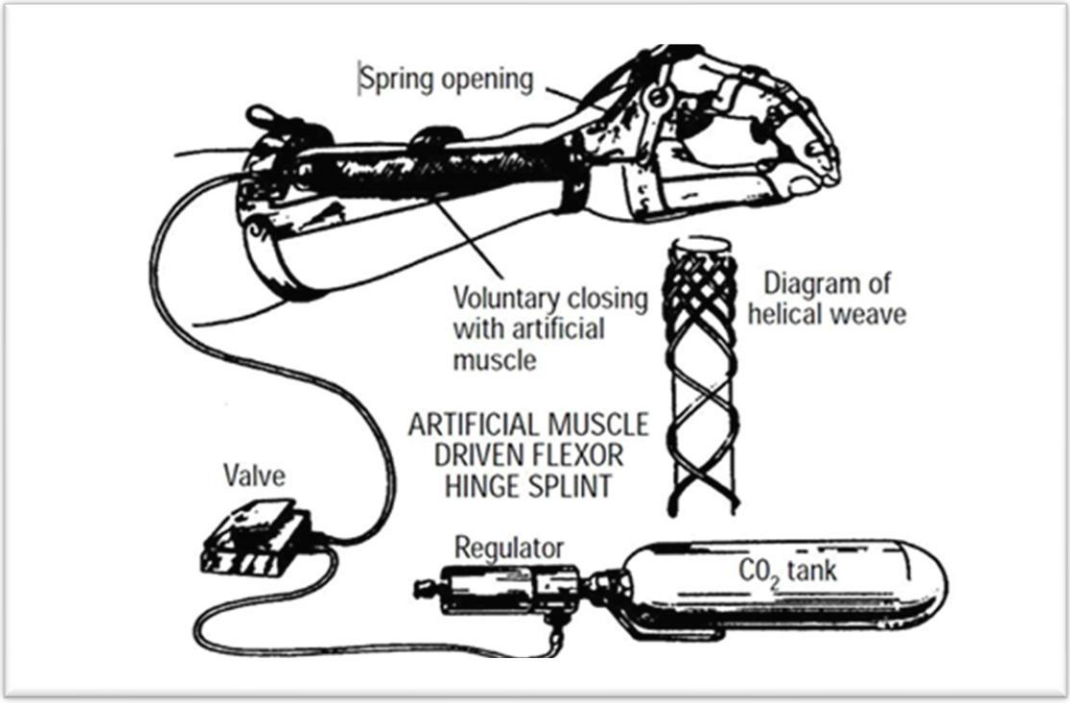


Figure 1: McKibben PAM Components (Tondu and Lopez, 1997)

In a first approach toward the model of a McKibben type PAM, losses are not considered and thus, the gas enters the PAM and expands, generating a traction force and a work. To generate the volume increase, the gas performs a work input (Win) that can be described.

$$dWin = \int (P - Pex)dli.dSi$$

$$dWin = (P - Pex) \int dli.dSi$$

$$dWin = P'dV$$

Where,

$Pex = External\ Force,$

$P = Absolute\ Pressure$

$P' = Relative/Gauge\ Pressure$

As the pressure performs work by volume expansion and it is assumed that the system has no loss, all the work is converted into traction and can be expressed by Wout,

$$dWout = -FdL$$

Where, F is the axial tension and dL is the displacement.

The Conservation of Energy gives:

$$dWin = dWout$$

$$P'dV = -FdL$$

$$F = -P' \frac{dV}{dL} \quad \text{..... (1)}$$

With the strength defined in terms of the volume change, just missing define the volume variation permitted by the mesh. The variation stops when volume reaches its maximum. At this moment, the muscle reaches the largest contraction.

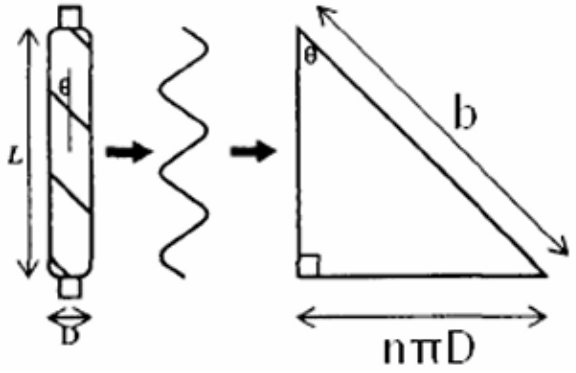


Figure 2: PAM Geometry (Tondu and Lopez, 1997)

Observing Fig.3, it is possible to make the length (L) and diameter (D) as a function of the angle (θ). This angle is one formed between the mesh and the axial length of the muscle. The length of fiber can be named b and n is the number of turns of the fiber around the muscle. Using b and n, it is possible to set the length and the diameter of the muscle.

$$cos\theta = \frac{L}{b} \quad \text{..... (1)}$$

$$L = b\,cos\theta \quad \text{..... (2)}$$

$$sin\theta = \frac{Dn\pi}{b}$$

$$D = \frac{bsin\theta}{n\pi} \quad \text{..... (3)}$$

Making the Volume equation as function of the diameter and D & length L.

$$V = \frac{1}{4} * \pi D^2 L \quad \text{..... (4)}$$

By substituting D & L in eq 4

We get,

$$V = \frac{b^3}{4\pi n^2} * [sin^2\theta * cos\theta] \quad \text{..... (5)}$$

Diff. eq (2) & eq (5) w. r. t. θ

$$\frac{dL}{d\theta} = -bsin\theta \quad \text{..... (6)}$$

$$\frac{dV}{d\theta} = \frac{b^3}{4\pi n^2} * [3cos^2\theta - 1] \quad \text{..... (7)}$$

Substitute the value of eq (6) & eq (7) in eq (1)

$$F = -P' * \frac{dV}{dL} = -P' * \frac{\frac{dV}{d\theta}}{\frac{dL}{d\theta}}$$

After substituting and rearranging, we get:

$$F = \frac{P'D^2\,(3cos^2\theta - 1)}{4\pi n^2} \quad \text{..... (8)}$$

$$D = \frac{bsin\theta}{n\pi}$$

$$\frac{D}{sin\theta} = \frac{b}{n\pi}$$

By substituting the value of D & trigonometric relation in eq (8)

$$F = \frac{\pi}{4} P'D^2 * \frac{(3cos^2\theta - 1)}{sin^2\theta} \text{ ..... (9)}$$

Adopting contraction (ε) and applying Eq.9, it is possible to reach to required formulation (Daerden and Lefeber, 2000).

$$D = \frac{bsin\theta}{n\pi} \text{ ..... (After applying pressure)}$$

$$D_0 = \frac{bsin\theta_0}{n\pi} \text{ ..... (Before applying pressure)}$$

$$D = D_0 \text{ ..... Basic Interpretation}$$

So, we can replace D & sinθ with D0 & sinθ$_0$ in eq (9)

$$cos\theta = \frac{L}{b}$$

$$L = b\, cos\theta$$

$$L_0 = b\, cos\theta_0$$

$$\frac{L}{L_0} = \frac{cos\theta}{cos\theta_0}$$

By Rearranging,

$$cos\theta = \frac{L}{L_0} cos\theta_0 \text{ ..... (10)}$$

Where, Contraction Factor (ε)

$$\varepsilon = \frac{L_0 - L}{L_0} = 1 - \frac{L}{L_0}$$

$$1 - \varepsilon = \frac{L}{L_0}$$

Put this value in eq (10)

$$cos\theta = (1 - \varepsilon) * cos\theta_0$$

Substitute cosθ, D = D$_o$ & sinθ = sinθ$_o$ in eq (9)

$$F = \frac{\pi}{4} P'D^2 * \frac{(3(1-\varepsilon)^2 * cos^2\theta_0 - 1)}{sin^2\theta_0} \text{ ..... (11)}$$

This Formula is for static conditions only.

By rearranging eq (11) and converting it into bar,

$$P' = \frac{F*4*sin^2\theta_o}{\pi*D^2*(3(1-\epsilon)^2*cos^2\theta_o - 1)*10^5} \text{ ..... (12)}$$

Losses:

There are various losses in calculating force, such as Braid Friction loss, Bladder elasticity loss, Hydraulic Pressure Loss, End Fitting and seal loss, and much more. But this are main losses that are considered.

$$F_{loss} = F_f + F_e + F_h + F_s$$

Where,

$F_f =$ Braid friction loss

$F_e =$ Bladder elasticity loss

$F_h =$ Hydraulic pressure loss

$F_s =$ End fitting and sealing loss

If we want to calculate the practical force which we have to require,

$$F_{practical} = \frac{P\,D_0^2}{4sin^2\theta_0} * (3(1-\varepsilon)^2 cos^2\theta\ - 1) - F_f - F_e - F_h - F_s$$

1. **Braid Friction loss:**

$$F_f = \mu_b * N_c$$

Where,

$\mu_b =$ Braid friction coefficient

$N_c =$ total normal contact force between the braid and bladder

2. **Bladder elasticity loss:**

$$F_e = EA\varepsilon$$

Where,

$E =$ Young's modulus of bladder

$A =$ Cross − sectional area of bladder wall

$\varepsilon =$ Axial strain

Area: $A = \frac{\pi}{4}\left(D_o^2 - D_i^2\right)$

Where, D_o and D_i are the outer and inner diameters.

3. **Hydraulic Pressure Loss:**

$$F_h = \Delta P A_i$$

Pressure drop:

$$\Delta P = f\frac{L_h}{D_h}\frac{\rho v^2}{2}$$

Where,

$f =$ Darcy friction factor

$L_h =$ Hose length

$D_h =$ Hose inner diameter

$\rho =$ Fluid density

$v =$ Fluid velocity

$A_i =$ Internal fluid area

Internal area:

$$A_i = \frac{\pi D_i^2}{4}$$

4. **End Fitting and seal loss:**

$$F_s = \mu_s N$$

Where,

$\mu_s = \text{Seal friction coefficient}$

$N = \text{Normal force on seal}$

Let us consider, by combining all the losses in force be 30%, which simplifies our further calculation and makes easy to calculate.

Calculations:

By using this Formula:

$$F = \frac{\pi}{4} P' D^2 * \frac{(3(1-\varepsilon)^2 * cos^2\theta_0 - 1)}{sin^2\theta_0}$$

We can get the Force and Pressure relation.

Actuator Parameters For test trial 1:

- Length of bladder: 300mm (complete length), 250mm (active length) (we can take any length as the calculation for 0 to infinity)

$$\boldsymbol{L = 120mm = 0.12m}$$

- Diameter of bladder: for trial we will go for 10mm

$$\boldsymbol{D = 10mm = 0.01m}$$

- Contraction factor: $\boldsymbol{\varepsilon = 0.2 \text{ to } 0.25}$
- Operating Pressure: **0 to 6 bar**
- Force That required to lift the minimum of 5kg of payload

$$\boldsymbol{F = 5 * 9.81 = 49.05\ N}$$

- So, considering the required force is 10 times of payload,

$$\boldsymbol{F = 10 * 5 * 9.81 = 490.5\ N}$$

- Now consider gear mechanism with gear ratio 1:2, we required 0.5 times of required force,

$$\boldsymbol{F = 0.5 * 490.5 = 245.25\ N}$$

  So, by considering Safety we will design for least 300N

After using this Formula:

$$F = \frac{\pi}{4} P' D^2 * \frac{(3(1-\varepsilon)^2 * cos^2\theta_0 - 1)}{sin^2\theta_0}$$

Pressure in Pascals: $1\ bar = P' * 10^5\ Pa$

We can determine the pressure value:

$$P' = \frac{F*4*sin^2\theta_o}{\pi*D^2*(3(1-\epsilon)^2*cos^2\theta_o - 1)*10^5}$$

On D = 0.01m, θo = 20° , ε = 0.2 and F = 300 N

$$\underline{\boldsymbol{P' = 6.63\ bar}}$$

$$\underline{\boldsymbol{P' = 6.63 * 10^5\ Pa}}$$

On D = 0.01m, θo = 20° , ε = 0.25 and F = 300 N

$$\underline{\boldsymbol{P' = 9.12\ bar}}$$

$$\underline{\boldsymbol{P' = 9.12 * 10^5\ Pa}}$$

On D = 0.01m, θo = 25° , ε = 0.2 and F = 300 N

$$\underline{\boldsymbol{P' = 11.82\ bar}}$$

$$\underline{\boldsymbol{P' = 11.82 * 10^5 Pa}}$$

On D = 0.01m, θo = 25° , ε = 0.25 and F = 300 N

$$\boldsymbol{P' = 17.67\ bar}$$

$$\boldsymbol{P' = 17.67 * 10^5\ Pa}$$

Considering efficiency of 30% of maximum obtained value,

$$\boldsymbol{P'max = 17.67 + 0.3 * 17.67}$$

$$\boldsymbol{P'max = 22.97 \approx 23\ bar}$$

$$\boldsymbol{P'max = 23 * 10^5\ Pa}$$

Now, calculate for losses and what should be maximum deflection that will required for rotation, and let us verify the considered 30% loss.

First, we use Modified Formula for HSA,

$$F_{practical} = \frac{P * 10^5 * D_0^2}{4sin^2\theta_0} * (3(1-\varepsilon)^2 cos^2\theta - 1) - (F_f - F_e - F_h - F_s)$$

We have P = 6.63, D_0 = 10 mm = 0.01 m, θ_0 = 20º, ε = 0.2

We will calculate for two different arrangements,

1. Nilon 6 Braiding and Silicone Rubber Tube
2. Aramid Braiding and Nitrile Butadiene Rubber (NBR) Tube

Material Properties:

Property / Dimensions	**Nilon 6**	**Aramid**
Braid Friction Co-efficient (μ_b)	0.25 to 0.45	0.14 to 0.40
	0.35	0.27
Diameter	16 mm	12.7 mm
Length	250 mm	250 mm

Property / Dimensions	**Silicone**	**NBR**
Young's Modulus (E)	50 MPa	2 – 5 MPa
	50 MPa	3.5 MPa
ID	10 mm	10 mm
OD	14 mm	12 mm
Length	250 mm	250 mm

Table 2: Material Properties

For Arrangement 1,

Theoretical Force,

$$F = \frac{\pi}{4} P' D^2 * 10^5 * \frac{(3(1-\varepsilon)^2 * cos^2\theta_0 - 1)}{sin^2\theta_0}$$

$$F = \frac{\pi}{4} * 6.63 * 10^5 * 0.01^2 * \frac{(3(1-0.2)^2 * cos^2(20) - 1)}{sin^2(20)}$$

$$F = 309.554\ N$$

1. Braid Friction Loss,

$$F_f = \mu_b N_c$$

$$F_f = 0.35 * 100$$

$$F_f = 35\ N$$

The value of N_c is taken as illustration; it would change as we measure the value in practical basis.

2. Bladder elasticity loss:

$$F_e = EA\varepsilon$$
$$A = \frac{\pi}{4}(D_o^2 - D_i^2)$$
$$A = \frac{\pi}{4}(0.014^2 - 0.01^2) = 7.539 * 10^{-5}\ m^2$$
$$F_e = 50 * 10^6 * 7.539 * 10^{-5} * 0.05$$
$$F_e = 188.475\ N$$

3. Hydraulic Pressure Loss:

$$F_h = \Delta P A_i$$

Uses Enklo 68 Oil:

Pipe inner diameter:	10 mm = 0.01 m
Volumetric flow rate:	0.5 L/min
Operating temperature:	40°C
Oil density:	890 kg/m³

Calculation:

$$V = \frac{Q}{A_i}$$
$$A_i = \frac{\pi D_i^2}{4} = \frac{\pi * 0.01^2}{4} = 7.854 * 10^{-5}\ m^2$$
$$V = \frac{0.5 * 10^{-3}}{60 * 7.854 * 10^{-5}} = 0.106\ m/s$$

$v = 68\ cSt$ at 40 °C

$$Re = \frac{VD}{v} = \frac{0.106 * 0.01}{68 * 10^{-6}} = 15.58$$

Flow is Laminar, Re < 2300

$$f_D = \frac{64}{Re} = \frac{68}{15.58} = 4.36$$
$$\Delta P = f\frac{L_h}{D_h}\frac{\rho v^2}{2}$$
$$\Delta P = \frac{4.36 * 0.25 * 890 * 0.106^2}{0.01 * 2} = 545\ Pa$$
$$F_h = 545 * 7.854 * 10^{-5} N$$
$$F_h = 0.0428N$$

4. End Fitting and seal loss:

Seal Friction Co-efficient: $\mu_s = 0.4$

Normal force on seal: $\mathrm{N} = 50\ \mathrm{N}$

$$F_s = \mu_s N$$
$$F_s = 0.4 * 50$$
$$F_s = 20\ N$$

Now, we will calculate Practical Force value

$$F_{practical} = F - F_f - F_e - F_h - F_s$$
$$F_{practical} = 309.554 - 35 - 188.475 - 0.0428 - 20$$
$$F_{practical} = 66.0112\ N$$

For Arrangement 1, we got

$$\boldsymbol{F_{practical} = 66.01\ N}$$

For Arrangement 2,

Theoretical Force,

$$F = \frac{\pi}{4} P' D^2 * 10^5 * \frac{(3(1-\varepsilon)^2 * cos^2\theta_0 - 1)}{sin^2\theta_0}$$
$$F = \frac{\pi}{4} * 6.63 * 10^5 * 0.01^2 * \frac{(3(1-0.2)^2 * cos^2(20) - 1)}{sin^2(20)}$$
$$F = 309.554\ N$$

1. Braid Friction Loss,

$$F_f = \mu_b N_c$$
$$F_f = 0.27 * 100$$
$$F_f = 27\ N$$

The value of N_c is taken as illustration, it would change as we measure the value in practical basis.

2. Bladder elasticity loss:

$$F_e = EA\varepsilon$$
$$A = \frac{\pi}{4}(D_o^2 - D_i^2)$$
$$A = \frac{\pi}{4}(0.012^2 - 0.01^2) = 3.4557 * 10^{-5}\ m^2$$
$$F_e = 3.5 * 10^6 * 3.4557 * 10^{-5} * 0.05$$
$$F_e = 6.047\ N$$

3. Hydraulic Pressure Loss:

$$F_h = \Delta P A_i$$

This value will be same as we are using hydraulic fluid,

$$F_h = 0.0428N$$

4. End Fitting and seal loss:

Seal Friction Co-efficient: $\mu_s = 0.4$

Normal force on seal: $\mathrm{N} = 50\ \mathrm{N}$

$$F_s = \mu_s N$$
$$F_s = 0.4 * 50$$
$$F_s = 20\ N$$

Now, we will calculate Practical Force value

$$F_{practical} = F - F_f - F_e - F_h - F_s$$
$$F_{practical} = 309.554 - 35 - 6.047 - 0.0428 - 20$$

$$F_{practical} = 248.464\ N$$

For Arrangement 2, we got

$$\boldsymbol{F_{practical} = 248.464\ N}$$

- **Rotating Mechanism calculations:**

  For this rotating mechanism, we will use Rack and Pinion mechanism, and one sector gear which consist arm of the hand. The pinion drives a larger spur gear, and the larger gear is attached to the arm carrying a 5 kg payload.

  Design inputs:

Payload mass	5 kg
Arm length	250 mm, assumed
Pitch diameters	Ø30 mm and Ø60 mm
Chosen Module for calculations	2 mm
Pressure angle	20°
Gear material	3D-printed PLA
Arm and gear orientation	Centre-line angle as 50° Arm angle needs separate treatment

Table 3: Gear Design Pre-assumptions

Min no of teeth:

$$z_{min} = \frac{2}{sin^2\varphi}$$

$$z_{min} = \frac{2}{sin^2 20} = 17.09 \approx 18\ teeths$$

$$so, z_p = 18\ and\ z_g = 36$$

Let us consider, the z_p = 18 and z_g = 36 (considering 1:2 teeth ratio)

The gear ratio:

$$i = \frac{z_g}{z_p} = \frac{36}{18}$$

$$i = 2:1$$

By using module (m) = 2 mm and pressure angle (ϕ) = 20°

Pitch diameter:

$$d = mz$$

For Pinion gear: $d_p = 2 * 18 = 36mm$

For Spur gear: $d_p = 2 * 36 = 72mm$

Outside diameter:

$$d_a = m(z + 2)$$

For Pinion gear: $d_p = 2(18 + 2) = 40mm$

For Spur gear: $d_p = 2(36 + 2) = 76mm$

Approximate root diameter:

$$d_f = m(z - 2.5)$$

For Pinion gear: $d_p = 2(18 - 2.5) = 31mm$

For Spur gear: $d_p = 2(36 - 2.5) = 67mm$

Base Diameter:

$$d_b = dcos(\varphi)$$

For Pinion gear: $d_p = 36 * \cos(20) = 33.83mm$

For Spur gear: $d_p = 72 * \cos(20) = 67.66mm$

Centre distance

For two external spur gears: $a = \frac{d_p + d_g}{2}$

$$a = \frac{36 + 72}{2} = 54mm$$

Circular Pitch: $p = \pi m$

$$p = \pi * 2 = 6.283\ mm$$

Reference tooth distance: $s = \frac{\pi m}{2}$

$$s = \frac{\pi * 2}{2} = 3.142\ mm$$

Addendum: $h_a = m = 2\ mm$

Dedendum: $h_f = 1.25 * m = 2.5\ mm$

Whole tooth depth: $h = 2.25 * m = 4.5\ mm$

Clearance: $c = 0.25 * m = 0.5\ mm$

Face width: $b = 10m = 10 * 2 = 20\ mm$

The rack geometry:

The rack and pinion must have the same module and pressure angle.

- Rack module: 2 mm
- Rack pressure angle: 20°

Calculate tooth dimensions

Rack pitch:

$$p = \pi m = 6.283mm$$

Theoretical tooth thickness at the pitch line:

$$s = \frac{\pi m}{2} = \frac{\pi * 2}{2} = 3.142mm$$

Addendum:

$$h_a = m = 2\ mm$$

Dedendum:

$$h_f = 1.25m = 1.25(2) = 2.5mm$$

Whole tooth depth:

$$h = h_a + h_f = 2 + 2.5 = 4.5\ mm$$

Theoretical bottom clearance:

$$c = 0.25m = 0.5\ mm$$

The nominal tooth thickness is measured at the pitch line. Do not use 3.142 mm as the width of the tooth at every height, because the tooth profile tapers.

Rack pitch-line position relative to the pinion, for 18-tooth pinion:

$$r_p = \frac{mz_p}{2} = \frac{2 * 18}{2} = 18\ mm$$

The rack pitch line must be 18 mm from the pinion center, measured perpendicular to the rack's direction of travel.

For a vertical rack, this means the horizontal distance from the pinion axis to the rack pitch line is 18 mm.

Important: The pitch line is a theoretical reference line inside the tooth geometry. It is not automatically the outside face or back face of the rack body.

Rack length and number of teeth

If the full 100 mm were toothed, the approximate number of pitches would be:

$$N \approx \frac{100}{6.283} = 15.92\ pitches$$

So, we can fit approximately 16 tooth pitches over 100 mm, depending on how we define the first and last tooth and the end margins.

Rack force and tooth load

Rack displacement is related to pinion rotation by:

$$\theta_p = \frac{s}{r_p} * \frac{180}{\pi}$$

The larger gear rotates at half the pinion's angular displacement.

Let, s = 1mm, meaning rack moves 1 mm

$$\theta_p = \frac{1 * 180}{18\pi} = 3.183°$$

Now, Rack moves 10 mm

$$\theta_p = \frac{10 * 180}{18\pi} = 31.83°$$

So, we can calculate by considering the gear ratio,

Rack movement	Pinion rotation	Spur gear rotation
1 mm	3.183°	1.592°
10 mm	31.83°	15.92°
25 mm	79.58°	39.79°

Table 4: Gear ratio and movement

These are theoretical values without backlash or elastic deformation.

Calculate the payload torque

The payload mass is 5 kg.

$$F = mg$$
$$F = 5(9.81) = 49.05\ N$$

Assuming the payload centre is 180 mm from the spur gear axis and the arm is horizontal:

$$T = FL$$
$$T = 49.05(0.18)\ = 8.829Nm$$

This is the static payload torque. It does not include the weight of the arm, acceleration, impact or friction.

Apply a service factor

For preliminary sizing, assume a service factor K_s =1.5

$$T_d = K_s T$$
$$T_d = 1.5(8.829) = 13.244\ Nm$$

Calculate the force on the gear teeth

Tangential force is:

$$F_t = \frac{T}{r}$$

For the driven spur gear:

$$F_t = \frac{13.244}{0.036} = 367.9N$$

The pinion torque for the ideal 2:1 ratio is half the spur gear torque:

$$T_p = \frac{13.244}{2} = 6.622Nm$$

The rack force is therefore:

$$F_{rack} = \frac{6.622}{0.018} = 367.9\ N$$

The tangential load is the same at the two ideal meshes. In a real mechanism, losses and dynamic effects can change the required actuator force.

Radial force and normal tooth force

For a pressure angle of 20°:

$$F_r = F_t \tan(\varphi)$$

$$F_r = 367.9 \tan(20) = 133.9\ N$$

The normal tooth contact force is:

$$F_n = \frac{F_t}{cos\varphi}$$

$$F_n = \frac{367.9}{\cos(20)} = 391.6N$$

The forces are:

Force	Calculated value
Tangential force	367.9 N
Radial separating force	133.9 N
Normal tooth contact force	391.6 N

Table 5: Forces

Select the face width:

Face width (b) is the width of the gear teeth along the gear axis.

A preliminary conventional gear guideline is:

$$b \approx 8(m)\ to\ 12(m)$$

For module 2 mm, that gives 16 to 24 mm. This is only a starting guideline. The final width must be determined from the strength calculation and the material.

Calculate tooth bending stress

Use the Lewis bending equation:

$$\sigma_b = \frac{F_t}{bmY}$$

Where:

- σ_b = nominal bending stress in MPa
- F_t= tangential force in N
- b = face width in mm
- m = module in mm
- Y = Lewis form factor

For preliminary calculations, use approximate 20° full-depth involute Lewis factors of 0.308 for an 18-tooth pinion, 0.376 for a 36-tooth gear, and 0.484 for a rack. Confirm the factor convention and exact tooth geometry before using these for a final design.

Pinion bending stress

$$\sigma_{b,p} = \frac{367.9}{12 * 2 * 0.308} = 49.7\ MPa$$

Spur gear bending stress

$$\sigma_{b,g} = \frac{367.9}{12 * 2 * 0.376} = 40.7\ MPa$$

Rack bending stress

$$\sigma_{b,r} = \frac{367.9}{12 * 2 * 0.484} = 31.6\ MPa$$

The pinion has the highest nominal bending stress in this comparison, so it is the first component to check.

These values are preliminary Lewis stresses. They do not include the full tooth-root stress concentration, print defects, fatigue, creep or misalignment.

Calculate the required face width

Rearrange the Lewis equation:

$$F_t = \sigma m b Y$$

For the pinion:

$$b_{req} = \frac{F_t}{\sigma_{allow} * m * Y}$$

The allowable stress must be based on testing the actual printed material and the intended loading conditions.

Illustrative allowable stress	Required pinion width
5 MPa	119.5 mm
10 MPa	59.7 mm
15 MPa	39.8 mm
20 MPa	29.9 mm

Table 6: Allowable stress vs required pinion width

These stress values are examples to show the calculation, not certified design limits for PLA. A 12 mm face width produces a nominal pinion stress of about 49.7 MPa. You should not assume a typical printed PLA gear can safely withstand that stress in repeated operation.
Design implication: you may need a wider face, a larger module, lower load, a different transmission arrangement, or a more suitable material. The allowable stress must be established first.

10. Calculate load capacity and factor of safety

The nominal Lewis tangential load capacity is:

$$F_{cap} = \sigma_{allow} b m Y$$

For the pinion with $b = 12\ mm$ and $m = 2$ mm. For a **3D-printed PLA gear with 20% infill**, the recommended safe **allowable stress** is 6 to 10 MPa (870 to 1,450 psi) for standard operation, $\sigma_{allow} = 10\ MPa$.

$$F_{cap} = 10 * 12 * 2 * 0308 = 73.2\ N$$

That is much lower than the 367.9 N design load.
The bending factor of safety is:

$$FOS = \frac{\sigma_{allow}}{\sigma_b}$$

For the pinion:

$$FOS = \frac{10}{49.7} = 0.2$$

Iterations:

Illustrative allowable stress	Pinion FOS
10 MPa	0.20
15 MPa	0.30
20 MPa	0.40
30 MPa	0.60
50 MPa	1.01

Table 7: Allowable stress vs FOS

These are illustrative calculations, not verified PLA material ratings. For an uncertain printed mechanism, a preliminary target of at least 2 against a verified, condition-appropriate allowable stress is a reasonable conservative starting point.
The rack pitch line should be tangent to the pinion pitch circle, with the pitch line 18 mm from the pinion axis.
Rack displacement is related to pinion rotation by:

Rack geometry:

The rack and pinion must have the same module and pressure angle.

Parameter	Symbol	Value
Module	m	2 mm
Pressure angle	ϕ	20°
Pinion teeth	z_p	18
Pinion pitch diameter	$d_p = mz_p$	36 mm
Pinion pitch radius	$r_p = \frac{d_p}{2}$	18 mm
Rack pitch	$p = m\pi$	6.283 mm
Rack tooth thickness at pitch line	$s = \frac{m\pi}{2}$	3.142 mm
Addendum	$h_a = m$	2 mm
Dedendum	$h_f = 1.25m$	2.5 mm
Whole tooth depth	$h = 2.25\ m$	4.5 mm
Bottom clearance	$c = 0.25m$	0.5 mm

Table 8: Rack Parameters

Required Rack movement to rotate the arm:

This is the equation to calculate the required rack travel for a desired arm rotation.
Since the spur gear radius is 36 mm:

$$s = r_g\theta_g$$

Here, θ_g must be in radians.
In degrees:

$$s = \frac{\pi d_g \theta_g}{360}$$

$$s = \frac{\pi * 72 * 10}{360}$$

$$s = 6.283\ mm$$

For your 72 mm pitch-diameter spur gear:

Desired arm rotation	Required rack travel
10°	6.28 mm
30°	18.85 mm
45°	28.27 mm
60°	37.70 mm
90°	56.55 mm

Table 9: Rack rotation

Braid Angle Calculation:

Initial Braid Angle:

$$\theta_0 = 20°$$

Formula to calculate Final Braid Angle:

$$cos\theta = (1 - \varepsilon) * cos\theta_0$$

Let Contraction Factor be calculated with,

$$\varepsilon = \frac{L_0 - L}{L_0}$$

We have,

Initial Length: L_0 = 250mm

And Contracted Length: L = 200mm

$$\varepsilon = \frac{250 - 200}{250} = 0.2$$

Now calculate Final Braid Angle:

$$cos\theta = (1 - \varepsilon) * cos\theta_0$$

$$cos\theta = (1 - 0.2) * \cos(20) = 0.7517$$

$$\theta = 41.26°$$

Similarly, we can calculate from angle 5° to 54°.

The things which have to keep in mind,

We want linear contraction and radial expansion.

If,

θ < 54.7° (Linear Contraction and radial expansion)

θ > 54.7° (Linear expansion and radial Contraction)

θ = 54.7° (Critical Value)

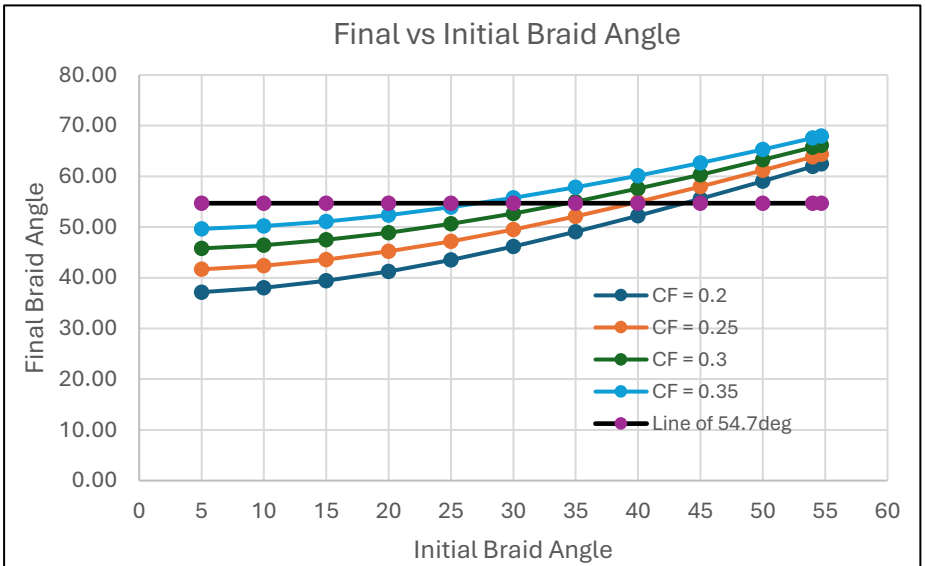
As we calculated for Initial Braid Angle 20° and Contraction Factor 0.2, there are further calculation in the table, and there is also a graphical representation of the angles and how they approach to the critical line which is 54.7° line.

Analytical Calculation:

Initial Braid Angle (Degree)	Final Braid Angle (Contraction Factor)			
	0.2	0.25	0.3	0.35
5	37.16	41.66	45.79	49.64
10	38.02	42.39	46.42	50.20
15	39.40	43.58	47.46	51.11
20	41.26	45.19	48.87	52.35
25	43.53	47.18	50.62	53.91
30	46.15	49.49	52.68	55.74
35	49.06	52.09	55.01	57.83
40	52.21	54.93	57.57	60.14
45	55.55	57.97	60.33	62.64
50	59.05	61.18	63.26	65.30
54	61.95	63.84	65.70	67.54
54.7	62.47	64.32	66.14	67.94

Table 10: Angle Calculations

Graphical Representation of Final Braid Angle vs Initial Braid Angle



Graph 1: Braid Angle Comparison

Conclusion:

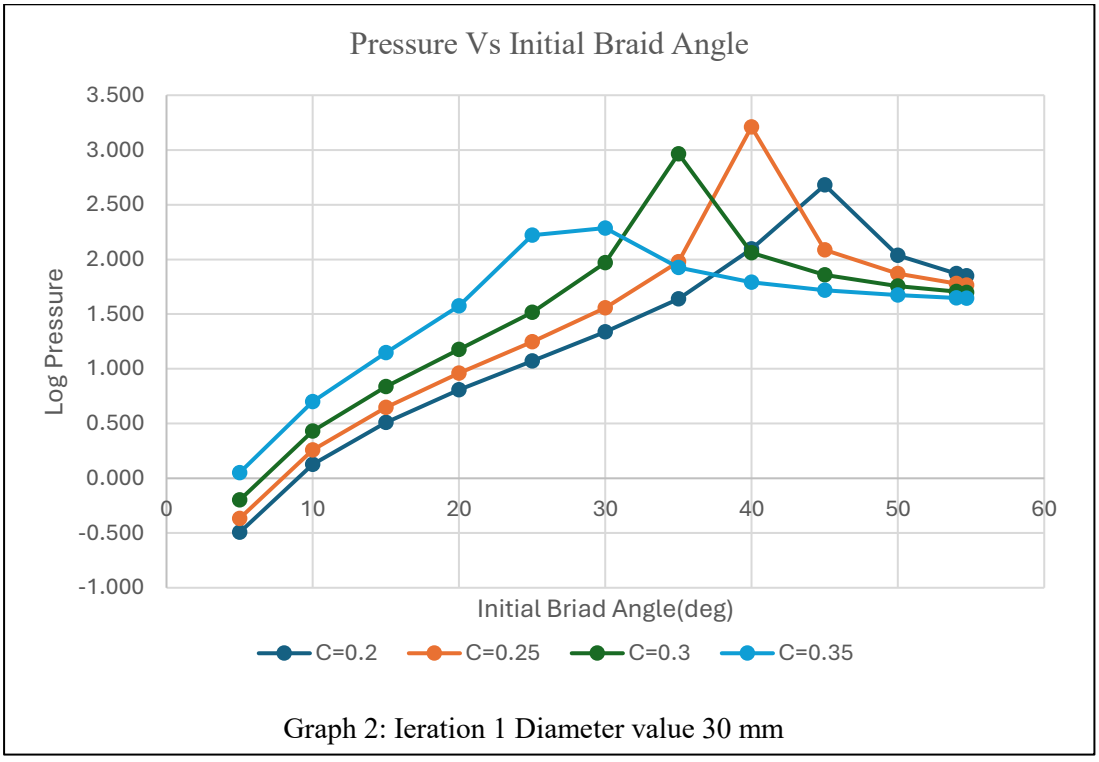
- Workable: 15º, 20º and 25º -> good response at $\varepsilon \leq 0.35$
- Can use 20º -> at $\varepsilon \leq 0.2$
- $\theta < 54.7°$ (Linear Contraction and radial expansion)
- $\theta > 54.7°$ (Linear expansion and radial Contraction)
- $\theta = 54.7°$ (Critical Value)

Different Iterations on diameter variation:

Iteration 1:

Initial Braid Angle (θ0) (Degree)	Pressure in bar (Contraction Ratio)				Pressure Value (Logarithmic) (Contraction Ratio)			
	0.2	0.25	0.3	0.35	0.2	0.25	0.3	0.35
5	0.320	0.430	0.632	1.125	-0.494	-0.366	-0.199	0.051
10	1.336	1.809	2.706	5.023	0.126	0.257	0.432	0.701
15	3.233	4.454	6.887	14.013	0.510	0.649	0.838	1.147
20	6.425	9.117	14.992	37.475	0.808	0.960	1.176	1.574
25	11.822	17.670	32.886	165.924	1.073	1.247	1.517	2.220
30	21.703	35.950	93.164	-193.403	1.337	1.556	1.969	2.286
35	43.582	94.964	-922.975	-84.060	1.639	1.978	2.965	1.925
40	124.561	-1621.285	-114.889	-61.601	2.095	3.210	2.060	1.790
45	-477.465	-122.231	-72.070	-52.146	2.679	2.087	1.858	1.717
50	-108.441	-74.034	-57.089	-47.061	2.035	1.869	1.757	1.673
54	-74.261	-59.955	-50.801	-44.478	1.871	1.778	1.706	1.648
54.7	-70.895	-58.286	-49.971	-44.113	1.851	1.766	1.699	1.645

Table 11: Initial Braid Angle vs Pressure

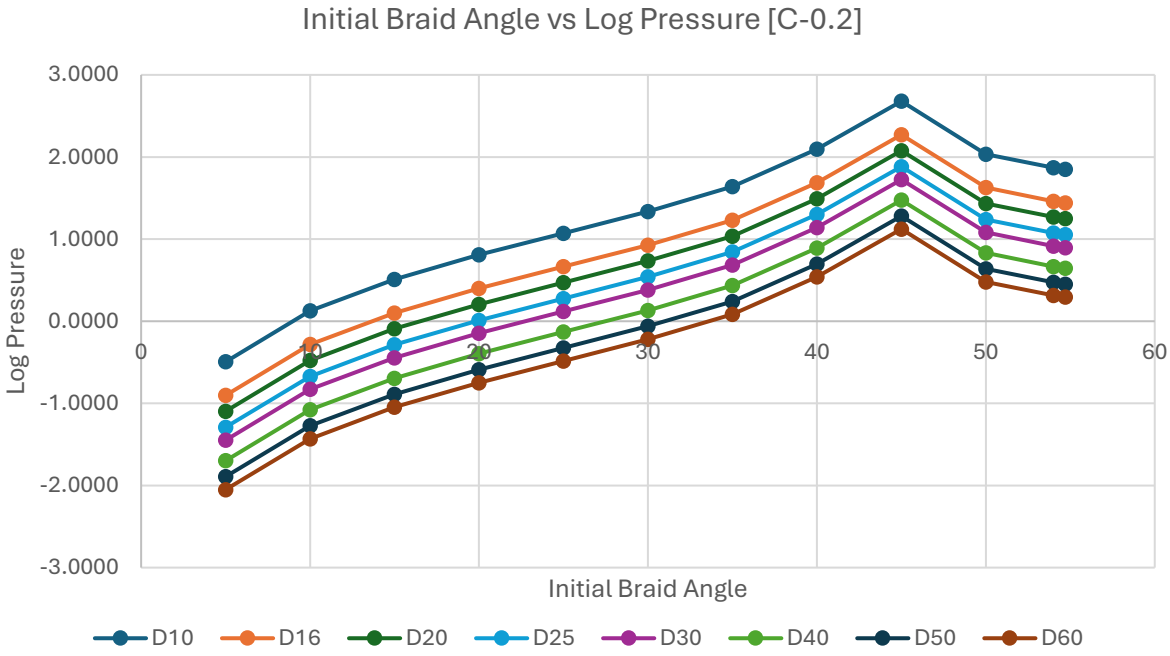


Graph 2: Ieration 1 Diameter value 30 mm

Conclusion:

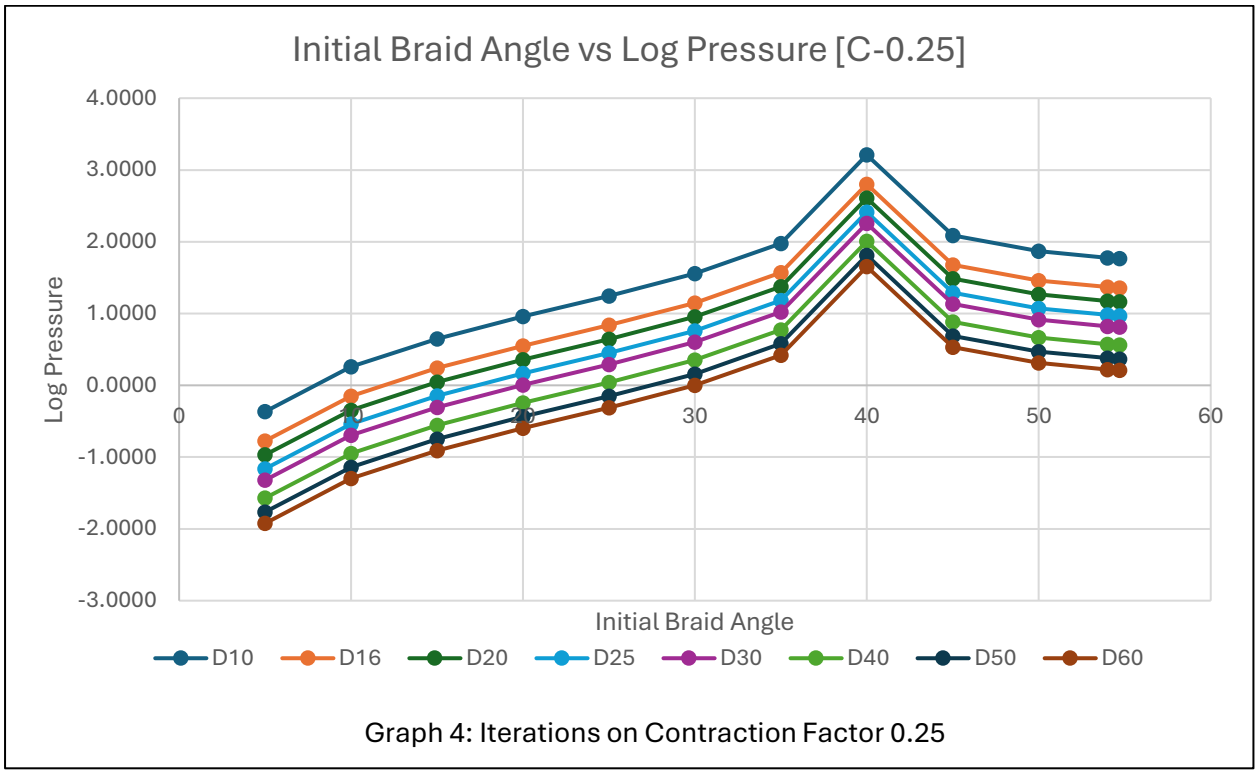
- Pressure becomes negative when the braid angle is 54.7° or more.
- The actuator works safely below 54.7°.
- A 25° initial braid angle is selected because it remains safe up to 35% contraction strain.
- The actuator shows linear contraction below 54.7° and tends to expand above 54.7°.
- Theoretical analysis was also carried out for actuator diameters of 16, 20, 25, 30, 40, 50, and 60 mm.
- By this way we can determine with multiple iterations.

Graphical Representation of Initial Braid Angle vs Logarithmic Value of Pressure on contraction factor 0.2 or 20%



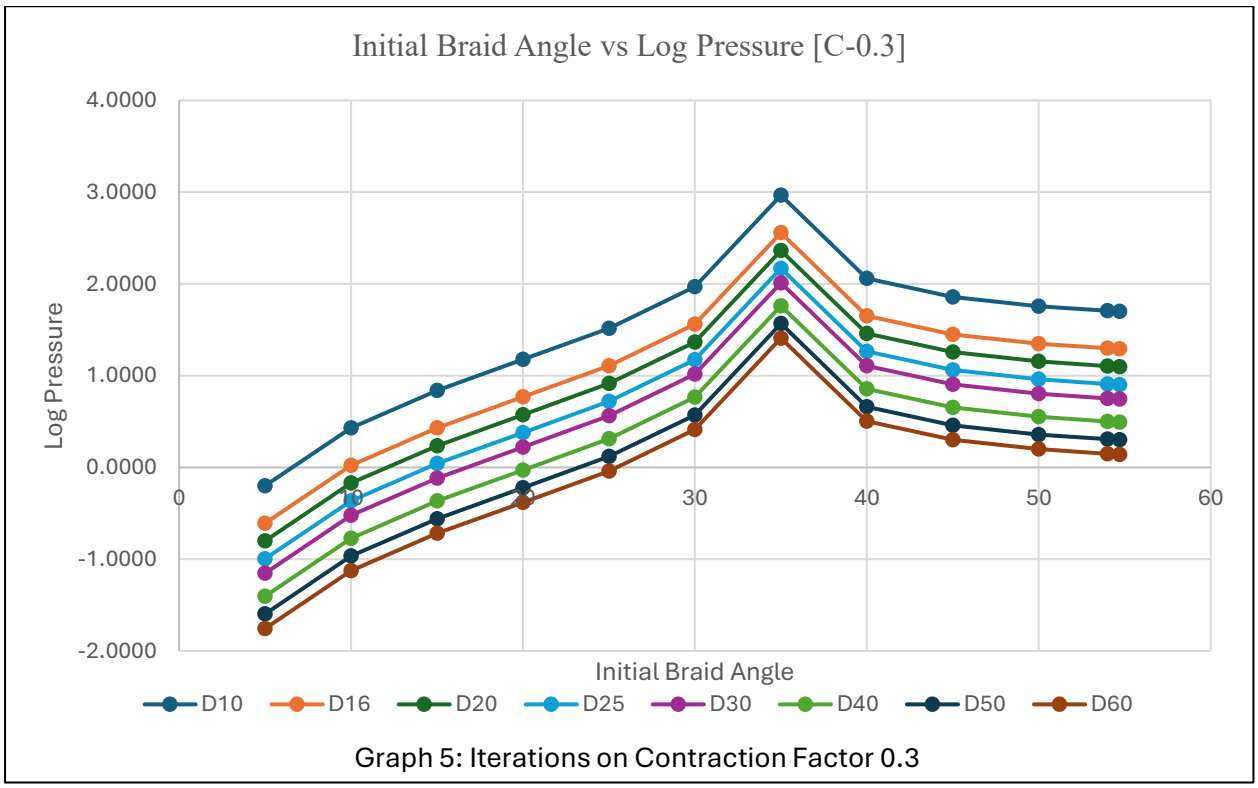
Graph 3: Iterations on Contraction Factor 0.2

Graphical Representation of Initial Braid Angle vs Logarithmic Value of Pressure on contraction factor 0.25 or 25%



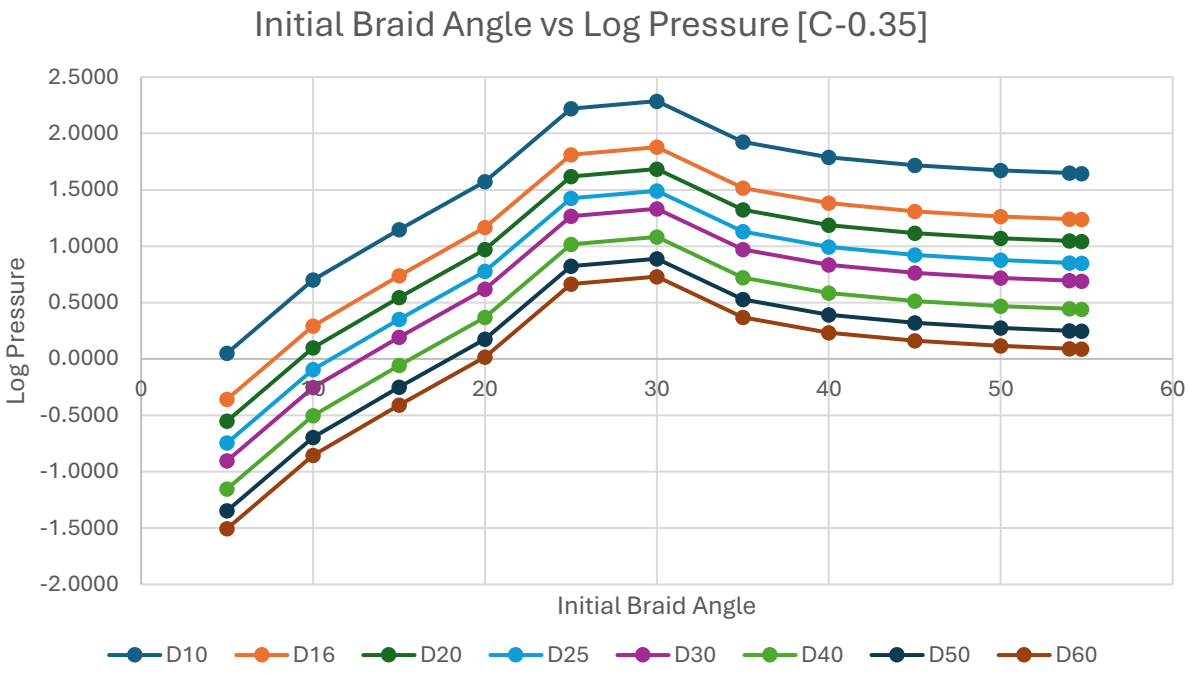
Graph 4: Iterations on Contraction Factor 0.25

Graphical Representation of Initial Braid Angle vs Logarithmic Value of Pressure on contraction factor 0.3 or 30%



Graph 5: Iterations on Contraction Factor 0.3

Graphical Representation of Initial Braid Angle vs Logarithmic Value of Pressure on contraction factor 0.35 or 35%



Graph 6: Iterations on Contraction Factor 0.35

Theoretical & Practical Pre-testing:

Formula

$$\Delta L_{th} = \varepsilon \times L_0$$

Given Data

- Active length, $L_0 = 120$ mm
- Contracted Length, L = 96 mm
- Contraction strain, $\varepsilon = 20\%$

Calculation

$$\Delta L_{th} = 0.2 \times 120 = 24 \text{ mm}$$

Practical Test Results:

Pressure (kg/cm^2)	NBR-Aramid (mm)	Silicone-Nylon (mm)
2	0	2
3	0	5
4.5	—	15
5	10	20
6	25	—
10	30	20

Table 12: Test results

1 kg/cm^2 = 0.98 bar

In practical pre-testing we have used two different rubber tubes and two different braiding. And paired with each other to obtain deflection under the zero load.

Rubber Tubes:

i. Silicon Rubber Tube (OD = 14mm, ID = 10mm)
ii. Nitrile Butadiene Rubber (NBR) Tube (OD = 12mm, ID = 10mm)

Braiding Material:

i. Nilon 6 (ID = 16mm)
ii. Aramid (ID = 12.7mm)

Configuration and arrangement used:

i. Silicon rubber tube with Nilon 6 Braiding
ii. NBR tube with Aramid Braiding

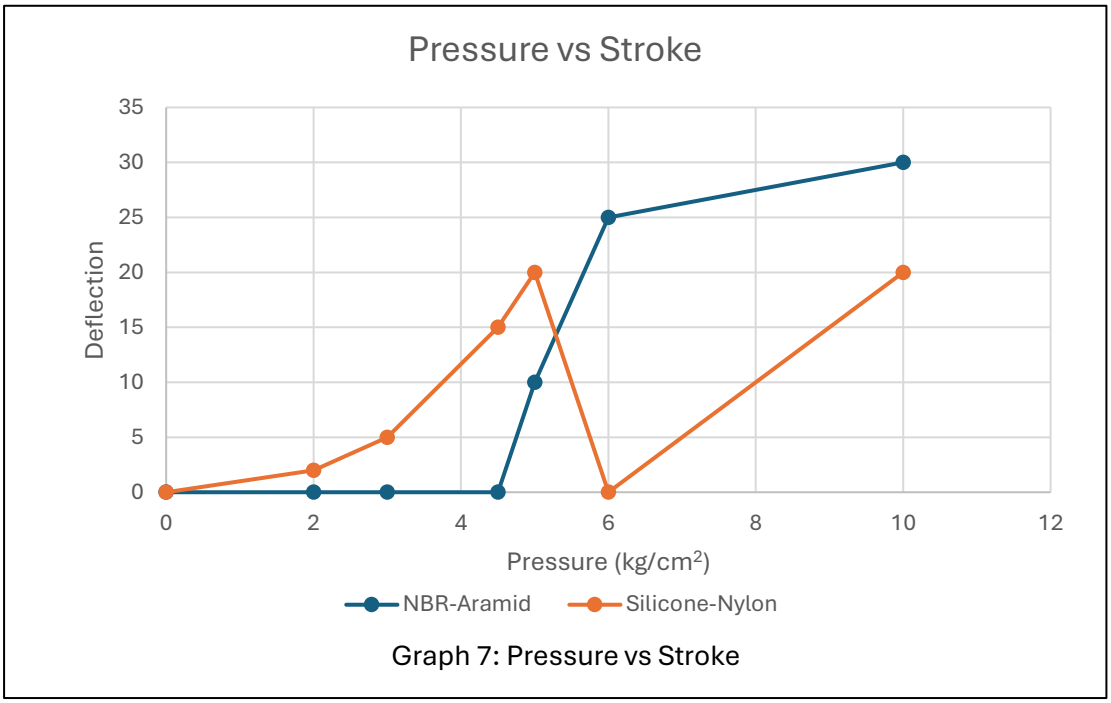
Fluid that used: HP Enklo 68



Before Contraction **After Contraction**

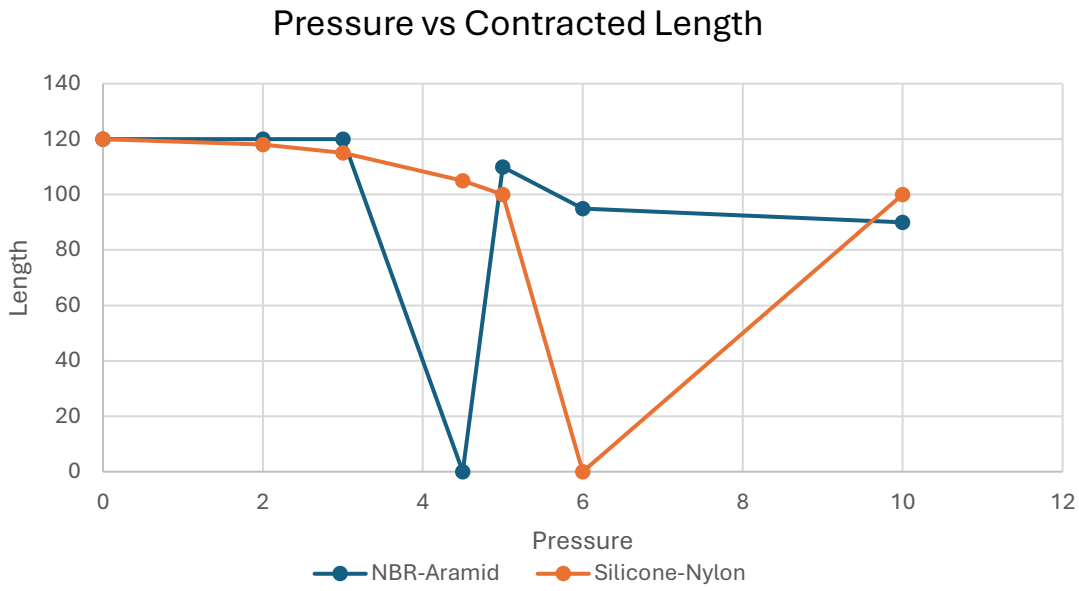
Figure 3: Image Before and After Contraction

Graphical Representation (Pressure vs Stroke):



Graph 7: Pressure vs Stroke

Graphical Representation (Pressure vs Contracted Length):

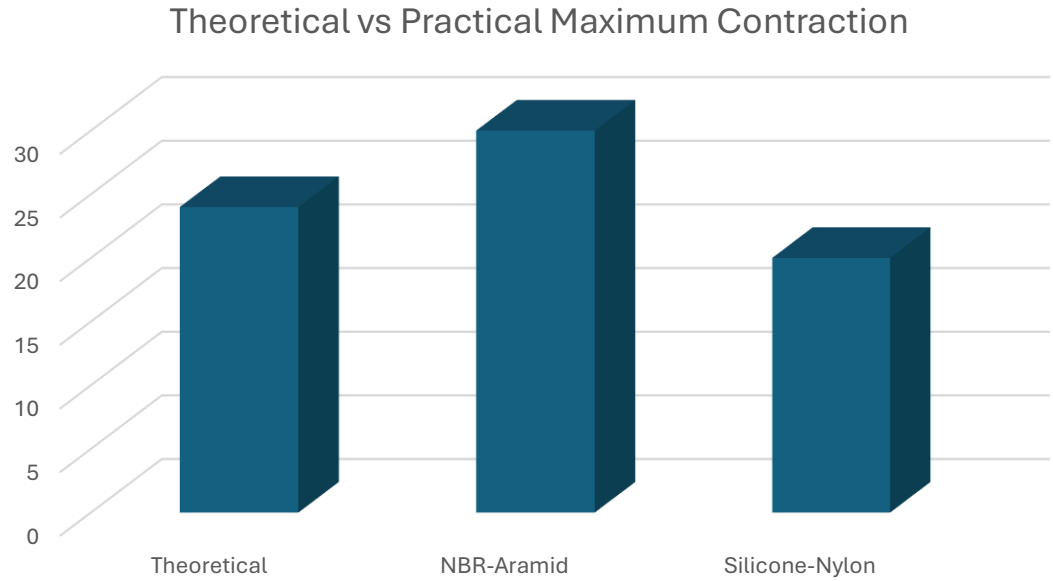


Graph 8: Pressure vs Contracted length

Maximum Length Change Comparison:

Type	Length Change (mm)
Theoretical	24
NBR-Aramid	30
Silicone-Nylon	20

Table 13: Length Comparison



Graph 9: Comparison of Contraction between theoretical and Practical Value

Images of Pre-testing:



Figure 4: Pre-Test set-up for Silicon Tube and Nylon Braiding



Figure 5: Pre-Test set-up for NBR Tube and Aramid Braiding

CAD Models:

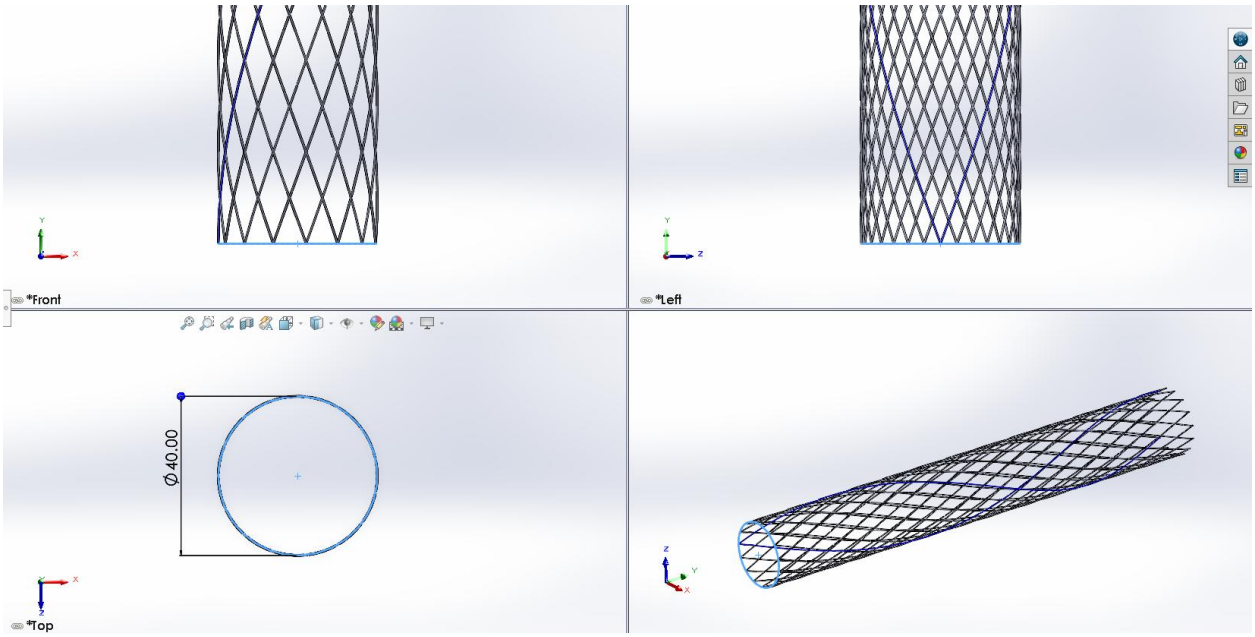


Figure 6: HSA Sheath

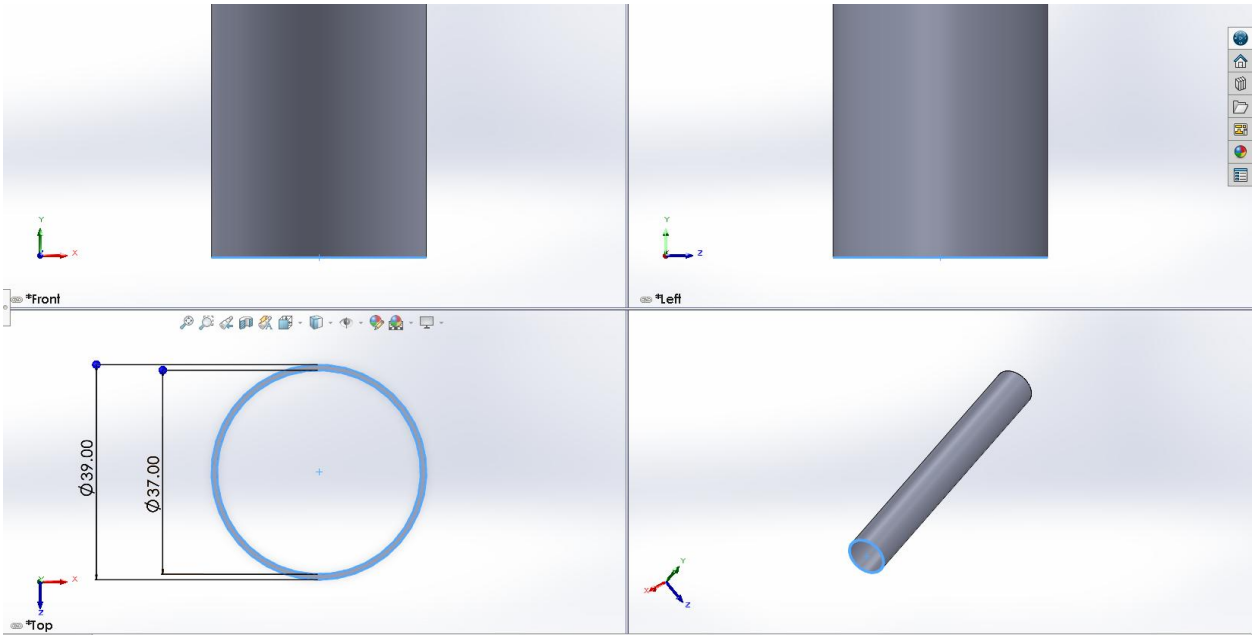


Figure 7: HSA Bladder

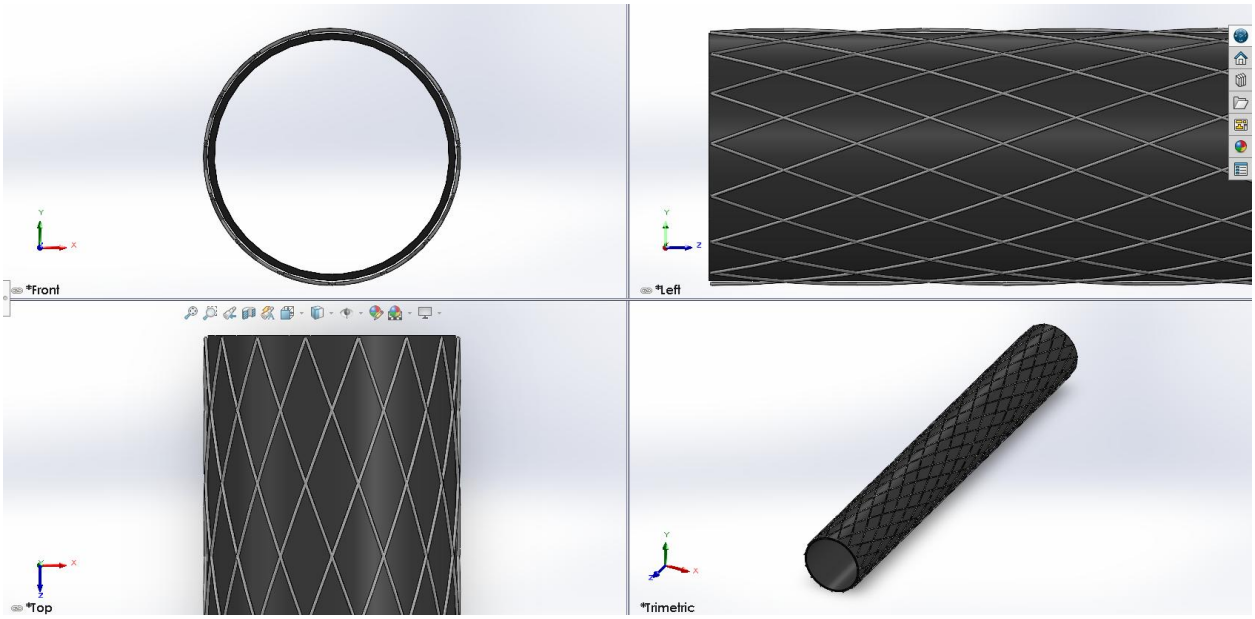


Figure 8: HSA Sheath and Bladder Mechanism

Rotating Mechanism for Arm:

We are planning to use rack and pinion mechanism, which a sector gear being a Forearm of the hand.

Further are some images of the gear mechanism which had done till now.

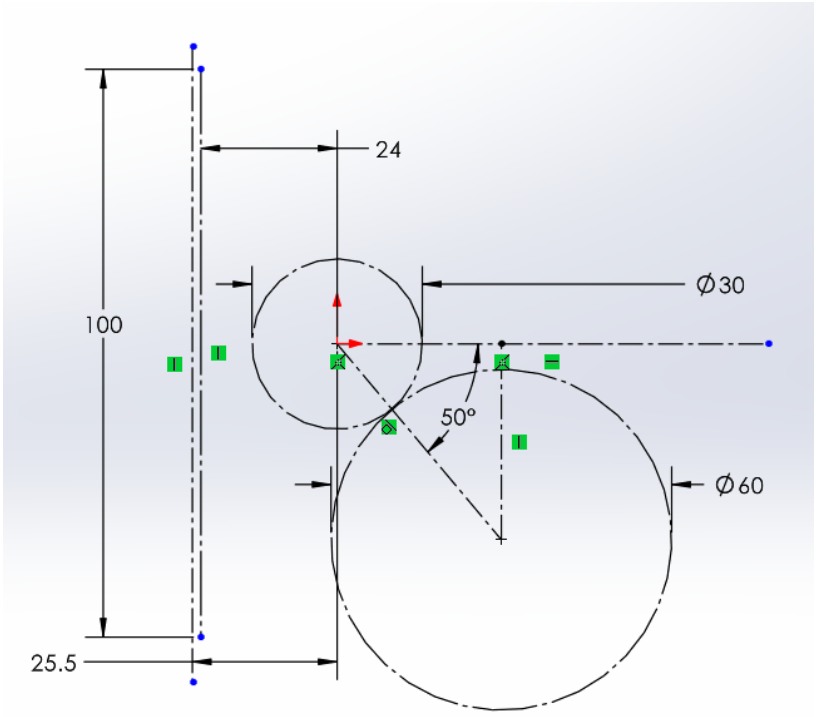


Figure 9: Reference Dimensions

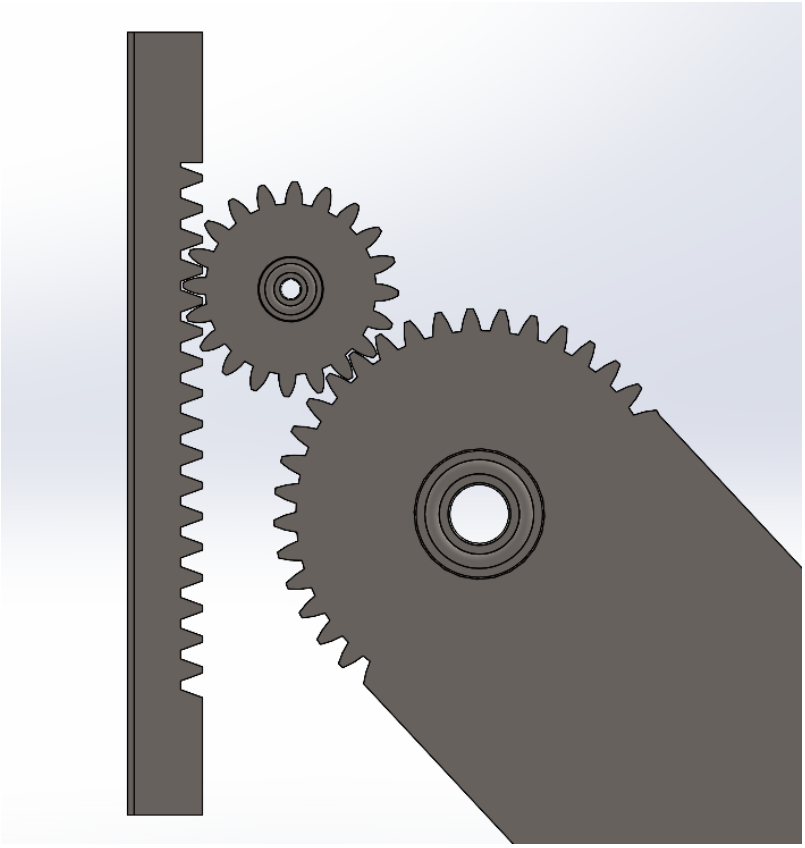


Figure 10: Sector Gear, Rack and Pinion arrangement

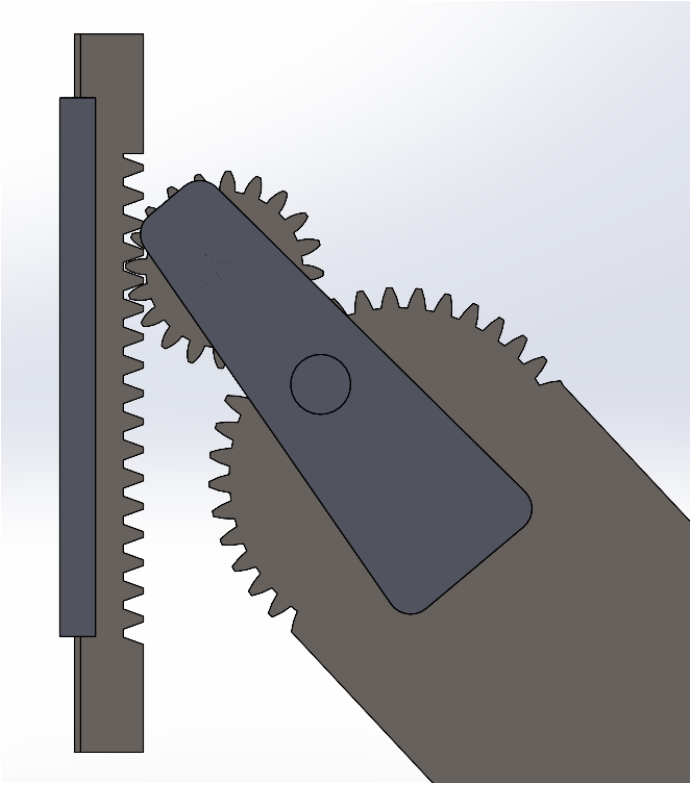


Figure 11: Image of guide and connector for gears and rack

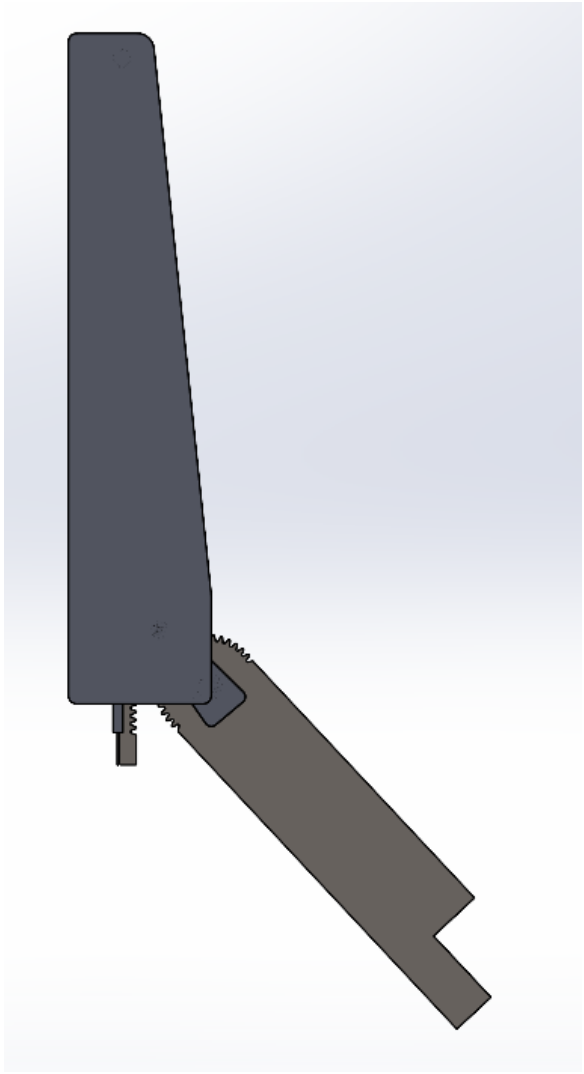


Figure 12: Complete arm mechanism



Figure 13: Reference from google

- This Kind of assembly we have to make:
  Like mostly of Biceps and triceps, using hydraulic (Water/Oil) as a working fluid for actuate the muscle.
- Hence forth the mechanism and hydraulic power pack we have to design requiring for this actuator.
- And enabling it's SDOF lifting.